

Section: Our Dynamic Universe

Topic: Forces, Energy and Power

A car of mass 1200 kg accelerates from rest along a straight, horizontal road. The engine exerts a constant force of 1800 N. The car travels 180 m in a time of 15 s. Air resistance and friction are negligible.

(a)
Calculate the acceleration of the car

Use:

$$s = ut + \frac{1}{2}at^2 \quad \text{with} \quad u = 0, \quad s = 180 \text{ m}, \quad t = 15 \text{ s}$$

$$180 = 0 + \frac{1}{2}a(15)^2 = \frac{1}{2}a(225) \Rightarrow a = \frac{180 \times 2}{225} = 1.6 \text{ m s}^{-2}$$

(b)
Calculate the final speed of the car

Use:

$$v = u + at = 0 + 1.6 \times 15 = 24 \text{ m s}^{-1}$$

(c)
Calculate the work done by the engine during this time

Work done = force × distance:

$$W = Fd = 1800 \times 180 = 3.2 \times 10^5 \text{ J}$$

(d)
Calculate the average power developed by the engine during this time

Power = work done / time:

$$P = \frac{W}{t} = \frac{3.2 \times 10^5}{15} = 2.1 \times 10^4 \text{ W} = 21 \text{ kW}$$

 Revision Tips

- For uniformly accelerated motion from rest, use:

$$s = \frac{1}{2}at^2 \quad \text{and} \quad v = at$$

- Work done is a **scalar**:

$$W = Fd \quad (\text{force and displacement must be in same direction})$$

- Power is rate of energy transfer:

$$P = \frac{W}{t}$$

- Always check units: power in watts, energy in joules, time in seconds

